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Unit Wise Question Bank

Department: Computer **Semester: I** **Academic Year: 2025 - 2026**

Class:SE **Div:** **Date:13/09/2025**

Course: Data Structure

Unit No. - name	Q. No.	Question	Marks	CO	Blooms Level
1-	Q.1	Define and explain the following terms: (i) Data (ii) Data structure (iii) Flowchart.	6	1	
	Q.2	Define algorithms and explain its characteristics	6	1	
	Q.3	Explain the divide and conquer strategy with suitable example. Comment on its time complexity.	6	1	
	.Q.4	Explain the Asymptotic notation Big O, Omega and Theta with suitable example	6	1	
	.Q.5	Explain static and dynamic data structures with examples.	6	1	
	Q.6	Differentiate between linear and non-linear data structure with example	6	1	
	Q.7	Explain the Greedy strategy with suitable example. Comment on its time complexity	6	1	
	Q.8	Define and explain ADT (Abstract Data Type).	6	1	
	Q.9	State Analysis of programming constructs - Linear, Quadratic, Cubic, Logarithmic	6	1	
	Q.10	. Define and explain the following terms: (a) Linear data structure (b) Non-linear data structure (c) Time complexity (d) Space complexity	6	1	

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Study material provided by: V. B. L

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	Q.11	Analyse time complexity of following codes. Show step count for each statement: [6]
		i) function is Prime(n){
		for (i=2;i<n;++i){
		if (n%i= = = 0){
		return false;
		}
		}
		return true;
		}
		
		ii) Function f()
		{
		ans=0
		for (i=n;i>=1;i/=2){
		for (j=1; j <=m; j*=2){
		ans+= (i*j)
		
		}
		print(ans)
		}
			6	1	

	Q.12	Write short note on. 1.ADT 2. Divide and conquer 3.greedy strategies. 4.array 5.space complexity. 6.Time complexity	6	1	
	Q.13	What is mean by algorithm? Explain the characteristics of algorithm	6	1	
	Q.14	Differentiate between **linear** and **non-linear** data structures with examples.	6	1	
	Q.15	What is the time complexity? why asymptotic notations are important?find time complexity of the following code using step count method.	6	1	

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		int x=0;
		for(i=0;i<N;i++)
		{
		for (j=N;j>i;j--)
		{
		x=x+i+j;
		}
		}			

	Q.16	Write short Note on: i) ADT ii) Divide and conquer strategy.	6	1	
	Q.17	Differentiate between: i) static and dynamic data structure Ii) Persistent and ephemeral data structure	6	1	
	Q.18	Explain step count method with the help of example.	6	1	

Course Teacher

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